20.3.2 Selection

Selection is one of the oldest methods of plant breeding in which individual plants or groups of plants are sorted out from a mixed population based on the morphological characters.

Methods of selection

There are three methods of selection. They are

1. Mass selection
2. Pureline selection
3. Clonal selection

1. Mass selection

Seeds of best plants showing desired characters are collected from a mixed population. The collected seeds are allowed to raise the second generation. This process is carried out for seven or eight generations. At the end, they will be multiplied and distributed to the farmers for cultivation.

Some common examples for mass selection are groundnut varieties like TMV–2 and AK–10. Its schematic representation is given below.

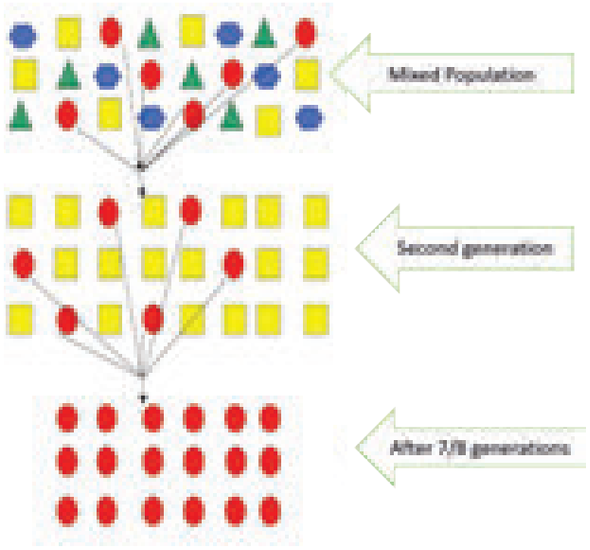


Figure 20.5 Mass Selection

2. Pureline selection

Pureline is "the progeny of a single individual obtained by self breeding". This is also called as individual plant selection. In pureline selection large numbers of plants are selected from a self-pollinated crop and harvested individually.

Individual plant progenies from them are evaluated separately. The best one is released as a pureline variety. Progeny is similar both genotypically and phenotypically.

3. Clonal selection

A group of plants produced from a single plant through vegetative or asexual reproduction are called **clones**.

All the plants of a clone are similar both in genotype and phenotype. Selection of desirable clones from the mixed population of vegetatively propagated crop is called **clonal selection**.

20.3.3 Polyploidy Breeding

Sexually reproducing organisms have two complete set of chromosomes in their somatic cells. This is called **diploid** (2n). The gametic cells have only one set of chromosome. This is called **haploid** (n). An organism having more than two sets of chromosomes is called **polyploid** (Greek : Polys = many + aploos = one fold + eidos = form). Such condition is called Polyploidy. It can be induced by **physical agents** such as heat or cold treatment, X-rays and **chemical agents** like colchicine.

Achievements of polyploidy breeding

Some achievements of polyploidy breeding are

a. Seedless watermelons (3n) and bananas (3n).
b. TV-29 (triploid variety of tea) with larger shoots and drought tolerance.
c. *Triticale* (6n) is a hybrid of wheat and rye. To make this plant fertile polyploidy is induced. It has higher dietary fibre and protein.
d. *Raphano brassica* is an allotetraploid by colchicine treatment.

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